

Study cohorts

PRIMARY (frozen checkpoint)
Einhaus 2023 OSCC
48 patients / 142 ROIs / 39 markers

SCCHN-COHORT CONTROL
IMMUcan, 2 samples
(cohort in pretraining corpus)
38 channels

CROSS-CANCER CONTEXT
5 IMC cohorts, all listed in
the 32-cohort corpus
(descriptive only, no
corpus-membership contrast)

Audit protocol

AUDIT PROTOCOL

ESM-2 marker embeddings
(39 markers, $\cos=1.00$
vs released)

uq0.99 clip + log1p
+ z-score

128×128 windows, top-2 density
per selected ROI (Task 2:
36/142 ROI deterministic
subsample)

provenance locked
(SHA-256, 3-way match)

TWO EVALUATION PITFALLS

label-derivation
circularity (probe vs
intensity baseline)

cluster seed-instability
(ARI 0.38-0.43)

Frozen-checkpoint tasks & outputs

TASK 1 Cell phenotyping
frozen-feature linear probe
patient-level GroupKFold(5)
macro-F1 = 0.713 vs
intensity baseline 0.783

TASK 2 Masked-channel
reconstruction
full-channel masking ×39
window-exchange permutation
+ ridge/template baselines

TASK 3 Token-space
phenotype recovery
UOP discovery / STA validation
multi-seed stability audit
(seed-unstable at K=20)

OUTPUT
capability map + audit checklist
+ full evidence package
(all numbers traceable)